MILITARY SPECIFICATION

HOSE ASSEMBLY

This specification is approved for use within the Army Missile Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification defines requirements for four types of hose assemblies.

1.2 Classification. The hose assemblies shall be of the following types (see 6.2).

TYPES

Type I - figure 1
Type II - figure 2
Type III - figure 3
Type IV - figure 4

1.3 Military part number. The military part number shall consist of the letter M, the basic number of this specification and dash numbers denoting type.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

FSC 4720

AMSC N/A

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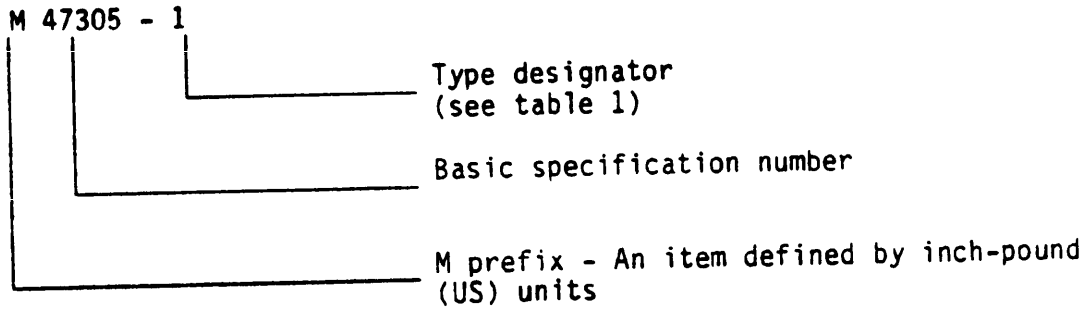


TABLE I. Type identification.

Type	Type Designator
I	-1
II	-2
III	-3
IV	-4

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-512 Plastic Sheet (Sheeting), Polyethylene

QQ-P-35 Passivation Treatments for Corrosion Resisting Steel

QQ-S-763 Steel Bars, Wire, Shapes and Forgings, Corrosion Resistant

PPP-B-636 Boxes, Shipping, Fiberboard

PPP-C-843 Cushioning Material, Cellulosic

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PPP-T-76	Tape, Packaging, Paper (For Carton Sealing)

MILITARY

MIL-B-117	Bags, Sleeve and Tubing - Interior Packaging
MIL-W-8611	Welding, Metal Arc and Gas, Steels and Corrosion and Heat Resistant Alloys, Process for
MIL-T-8808	Tubing, Steel, Corrosion Resistant (18-8 Stabilized) Aircraft Hydraulic Quality
MIL-H-25579	Hose Assembly Tetrafluoroethylene, High Temperature, Medium Pressure
MIL-P-27401	Propellant Pressurizing Agent, Nitrogen
MIL-P-55010	Plastic Sheet, Polyethylene Terephthalate
MIL-C-81302	Cleaning Compound, Solvent, Trichlorotrifluoroethane

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specification, Order of Precedence for the Selection of
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-45662	Calibration Systems Requirements
AN818	Nut, Tube Coupling, Short
MS33584	Tubing End - Standard Dimensions for Flared

(Copies of specification and standards required by contractors in connection with acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

SOCIETY OF AUTOMOTIVE ENGINEERS INC. (SAE)

AMS 5510	Steel Sheet, Strip and Plate, Corrosion and Heat Resistant 18Cr - 10.5Ni - 0.4Ti (SAE 30321), Solution Heat Treated
AMS 5512	Steel Sheet, Strip and Plate, Corrosion and Heat Resistant 18Cr - 10.5Ni - 0.80(Cb + Ta) (SAE 30347), Solution Heat Treated
AMS 5643	Steel Bars, Forgings, Tubing and Rings, Corrosion Resistant 16Cr - 4.0Ni - 0.30(Cb + Ta) - 4.0Cu, Solution Heat Treated
AMS 5687	Alloy Wire, Corrosion and Heat Resistant 73Ni - 15.5Cr - 8.0Fe Annealed
AMS 5689	Steel Wire, Corrosion and Heat Resistant 18Cr - 9.5Ni - 0.40Ti (SAE 30321), Solution Heat Treated
AMS 5697	Steel Wire, Corrosion Resistant 19Cr - 9.5Ni (SAE 30304), Solution Heat Treated
ARP 598	The Determination of Particulate Contamination in Liquids by the Particle Count Method

(Application for copies of SAE publications should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA, 15096.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification,

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however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2).

3.2 Materials. Specification and standards for necessary commodities and services not specified herein shall be selected in accordance with the order of precedence specified in MIL-STD-143.

3.2.1 Inner core. Inner core shall be flexible virgin tetrafluoroethylene conforming to MIL-H-25579, for use in a temperature range of -54°Celsius(C) to 93.3°C [-65°Fahrenheit(F) to 200°F].

3.2.2 Braid. The braid shall be corrosion resistant steel wire conforming to AMS 5689, AMS 5697 or AMS 5687 used in the cold drawn condition. There shall be no missing, kinked, broken or spliced strands of wire.

3.2.3 Components. With the exception of the material specified in 3.2.1 and 3.2.2 all parts shall be class 303, 304, 305 or stabilized 321 and 347 corrosion resistant steel in accordance with QQ-S-763, AMS 5510, and AMS 5512 or 17 - 4 PH in accordance with AMS 5643. Tubing shall be in accordance with MIL-T-8808. Welded components shall be class 321 or 347.

3.2.4 Finish. The detail parts and subassemblies, with the exception of the inner core (see 3.2.1), shall be passivated in accordance with QQ-P-35. Welds made after assembly of inner core and braid to the hose assembly shall be swab passivated.

3.2.4.1 Swab passivation. Cleaned parts shall be passivated by using a cotton swab at 10 minute intervals (total of six applications) for a minimum of 60 minutes using a solution in accordance with QQ-P-35 type II. At the conclusion of the passivation process, the parts will be rinsed in accordance with QQ-P-35.

3.3 Design and construction. The design and construction of the hose assembly shall conform to figure 1 for type I, figure 2 for type II, figure 3 for type III and figure 4 for type IV (see 4.6.1).

3.3.1 Dimensions. Dimensions and tolerances of the hose assembly shall be in accordance with the applicable figure for the type specified. Bend radii shall be 0.75 inch unless otherwise specified (see 4.6.1).

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3.3.2 Hose diameter. The hose assembly internal diameter consistency shall be not less than 0.560 inches and shall pass a 0.560 + 0.010 - 0.000 inch diameter ball freely with no restriction (see 4.6.1).

3.3.3 Fusion welding. All fusion welds shall be considered critical as defined by MIL-W-8611. Fusion welding shall be performed by the inert gas shielded nonconsumable electrode electric arc process and shall be in conformance with MIL-W-8611 except that radiographic inspection shall not be required.

3.3.4 Tube end. Tube ends shall be in accordance with MS 33584 with end fitting in accordance with AN818-12J or AN818-12K.

3.4 Performance characteristics.

3.4.1 Environmental. The hose assembly shall meet the requirements of this specification while being subjected to the following operating conditions and any combination thereof (see 4.6.2).

3.4.1.1 Temperature. Operational temperature range shall be -40°C to 93.3°C (-40°F to 200°F), (see 4.6.2.1 and 4.6.2.2).

3.4.1.2 Vibration. The hose assembly shall withstand vibration for 1.0 minute along each of three axes in accordance with figure 5 (see 4.6.2.3).

3.4.1.3 Shock. The hose assembly shall withstand shock up to 600 gravity units (g) for 1.25 ± 0.25 milliseconds duration (see 4.6.2.4).

3.4.2 Proof pressure. When tested in accordance with 4.6.3, the hose shall withstand 3390 pounds per square inch gage (psig) without leakage of 10 cubic centimeters per hour maximum of the tetrafluoroethylene inner core. There shall be no distortion (see 4.6.3).

3.4.3 Pressure impulse. When tested in accordance with 4.6.4, there shall be no evidence of leakage, fatigue cracks or rupture of more than 10 percent of the braid wire (see 4.6.4).

3.4.4 Burst pressure. When tested in accordance with 4.6.5, there shall be no bursting or breaking at any seam or junction (see 4.6.5).

3.5 Hose assembly cleanliness. The hose assembly shall be free of dirt, grease, oil, weld slag, scale and there shall be no particle contamination greater than specified in table II (see 4.6.6).

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TABLE II. Hose assembly contamination limitation.

Size Range (microns)	Number of particles (maximum allowable per 100 ml of fluid passed thru line)
<10	Unlimited
10 - 25	500
26 - 50	100
51 - 100	18
100 - 300	6
>300	None
Fibers 1/	None

1/ Fibers are defined as any particle whose length to diameter ratio exceed 10 to 1 regardless of composition. Fibers are counted as particles unless their length exceeds 300 microns.

3.6 Part number of interchangeable parts. All parts having the same manufacturers part number shall be functionally and physically interchangeable. Identification and part numbering requirements of DOD-STD-100 shall govern manufacturer's part numbering and changes thereto.

3.7 Identification or product. The following information shall be electro-chemically etched on each assembly in accordance with MIL-STD-130. Wire-on metal tags shall not be used.

Do not bend flexsection of hose under 6 inch minimum bend radius
Manufacturer's serial number
Manufacturer's code
Military part number
Proof pressure (inspectors stamp)

3.8 Workmanship. The hose assembly shall be constructed and finished in a thoroughly workmanlike manner. There shall be no cracks, breaks, burrs, dents, loose attaching parts or other evidence of poor workmanship that would make the hose assembly unsuitable for the purpose intended. Weld discoloration is permissible provided there is no scale or rust (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contact. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Inspection conditions shall be as specified in 4.6.

4.3.1 Equipment calibration. Equipment calibration shall be controlled in accordance with MIL-STD-45662.

4.4 First article inspection. First article inspection shall consist of all of the examinations and tests as specified under 4.6, and shall be conducted in the sequence of 4.6. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.2).

4.4.1 First article sample. The first article sample shall consist of one hose assembly of each required type, representative of the production product. The first article sample shall be subjected to the inspections and tests specified in 4.4.

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4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling (see 6.2) shall be in accordance with MIL-STD-105. An AQL of 1.0 percent shall be used for major characteristics and an AQL of 4.0 percent shall be used for minor characteristics in accordance with MIL-STD-105. When a sampling plan has been selected, it shall be utilized for the balance of procurement unless a change in sampling plans is approved by the procuring activity.

4.5.2 Quality conformance inspection. Utilizing the sampling plan of 4.5.1, the quality conformance inspection shall consist of the inspections and tests specified in table III.

TABLE III. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product------	3.3, 3.3.1, 3.3.2, 3.7 & 3.8	4.6.1
Proof pressure----------	3.4.2	4.6.3
Cleanliness----------	3.5	4.6.6

4.5.3 Noncompliance. Failure of any sample shall be cause for rejection of the entire lot represented by that sample as established in 4.5.1.

4.5.4 Inspection of packaging. The preservation, packing, packaging and marking shall be inspected to determine conformance with section 5.

4.6 Methods of inspection.

4.6.1 Examination of product. The hose assemblies shall be visually examined for conformance to the requirements of this specification. Individual units shall be examined for accuracy of dimensions, quality of workmanship and use of materials and finishes. The hose assembly internal diameter shall be checked for consistency (see 3.3, 3.3.1, 3.3.2, 3.7 and 3.8).

4.6.2 Environmental.

4.6.2.1 Low temperature. The hose assembly shall be tested in accordance with MIL-STD-810, method 502, procedure I and inspected for conformance with the requirements of 3.4.1 and 3.4.1.1.

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4.6.2.2 High temperature. The hose assembly shall be tested in accordance with MIL-STD-810, method 501, procedure I and inspected for conformance with the requirements of 3.4.1 and 3.4.1.1.

4.6.2.3 Vibration. The hose assembly shall be tested in accordance with MIL-STD-810, method 514, category V and inspected for conformance with the requirements of 3.4.1 and 3.4.1.2.

4.6.2.4 Shock. The hose assembly shall be tested in accordance with MIL-STD-810, method 516, procedure I and inspected for conformance with the requirements of 3.4.1 and 3.4.1.3.

4.6.3 Proof pressure. With the hose assembly in the restrained position as shown in the applicable figure and submerged in water, the hose assembly shall be pressurized to 3390 psig using nitrogen conforming to MIL-P-27401 for three minutes without leakage of 10 cubic centimeters per hour maximum leakage of the tetrafluroethelene inner core. Units shall be examined for leakage and distortion. The interior of the hose shall be kept dry during the test. At the end of testing the hose assembly shall be air dried at 200° ± 15°F without compromising the cleanliness requirements of 3.5 (see 3.4.2).

4.6.4 Pressure impulse. With the hose assembly in the restrained position as shown in the applicable figure the hose assembly shall be hydrostatically pressurized and bled of all air. The hose assembly shall be subjected to pressures from 0 to 3390 psig in 2 seconds and returned to 0 in 2 seconds for 300 cycles. Units shall be examined for leakage, fatigue, cracks or rupture of more than 10 percent of the braid wire (see 3.4.3).

4.6.5 Burst pressure. With the hose assembly in the restrained position as shown in the applicable figure the hose assembly shall be hydrostatically pressurized at a rate of 1000 psi per minute to 5090 psig. Units shall be examined for failure of any seam or junction (see 3.4.4).

4.6.6 Hose assembly cleanliness. Inspection of cleaned components shall be performed by the solvent rinse method. The solvent rinse shall be performed with filtered trichoroethylene conforming to MIL-C-81302, demineralized water or equivalent. A measured amount of 200 milliliters of fluid rinse per square foot of component inner (or effective) surface shall be used. Particle counting shall be in accordance with ARP 598 except that the entire sample collected per square foot of critical surface area shall be filtered. Particles shall not exceed the appropriate size ranges specified in 3.5.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified (see 6.2).

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5.1.1 Cleaning. Hose assemblies shall be cleaned in accordance with 3.5 and inspected in accordance with 4.6.6.

5.1.2 Unit packs. Each hose assembly shall be individually unit packed in a precleaned enclosure made from polyethylene terephthalate conforming to MIL-P-55010 and sealed. The unit pack shall be marked "Do not open until ready for use."

5.1.3 Intermediate packs. Each unit pack shall be individually packed in a precleaned intermediate enclosure made from polyethylene conforming to L-P-512 and sealed.

5.2. Packing. Hose assemblies preserved as specified in 5.1 shall be packed in fiberboard containers conforming to PPP-B-636 style RSC, type CF, grades 175, class DOM and cushioned with material conforming to PPP-C-843 type II. Container closure shall be in accordance with PPP-T-76 with barrier in accordance with MIL-B-117 style 2, type I, grade 4MIL, class B (see 6.2).

6. NOTES

6.1 Intended use. The hose assemblies are intended to be used to supply fluids in military systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Part number (see 1.3).

c. Whether first article inspection is required (see 3.1 and 4.4).

d. Sampling plan, if other than specified (see 4.5.1).

e. Preservation and packaging if other than specified (see section 5).

6.2.1.1 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first article. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the

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Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Subject term (key word) listing.

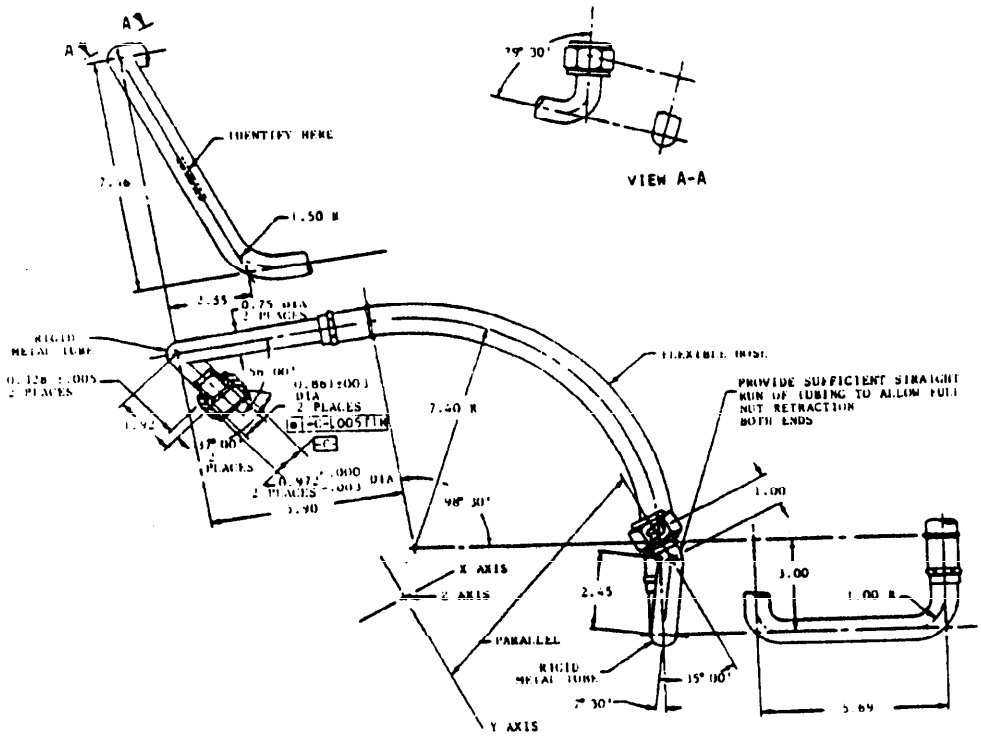
Hose assembly

Custodian:
Army - MI

Preparing activity:
Army - MI

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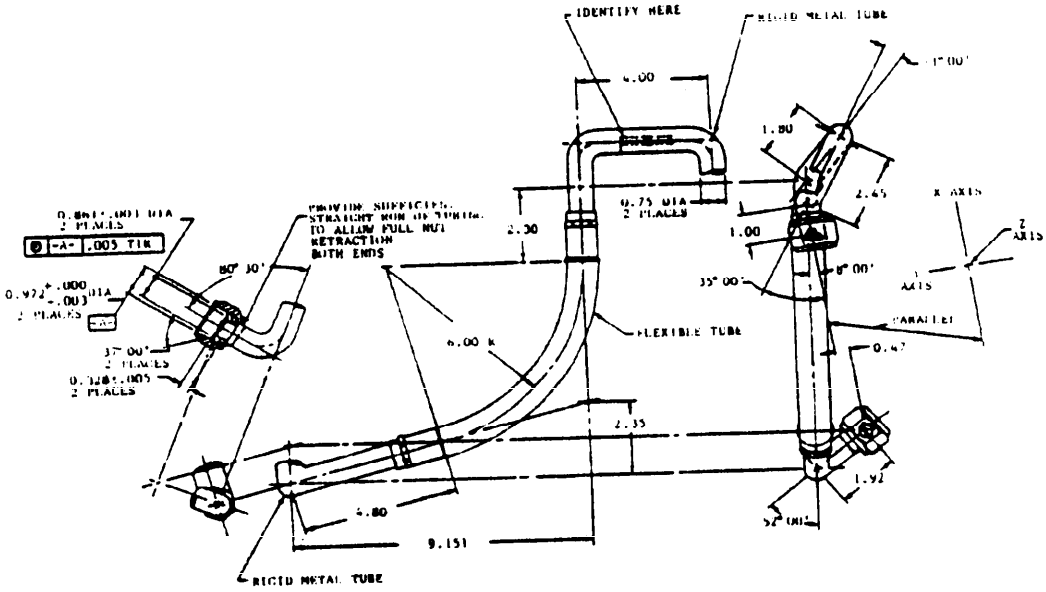


NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 1. Hose assembly type I.

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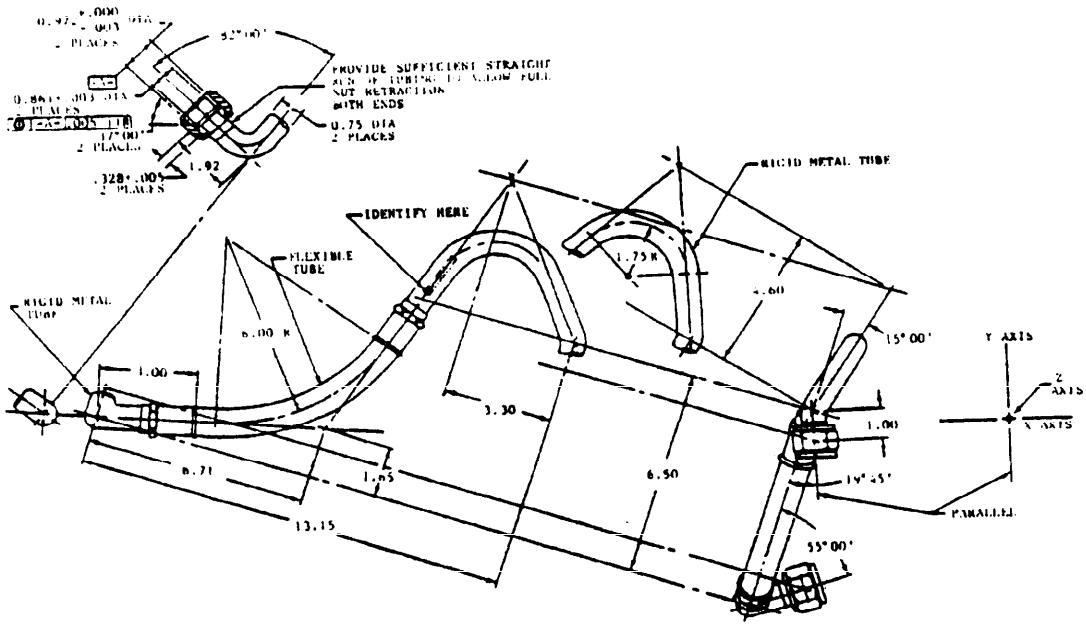


NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 2. Hose assembly type II.

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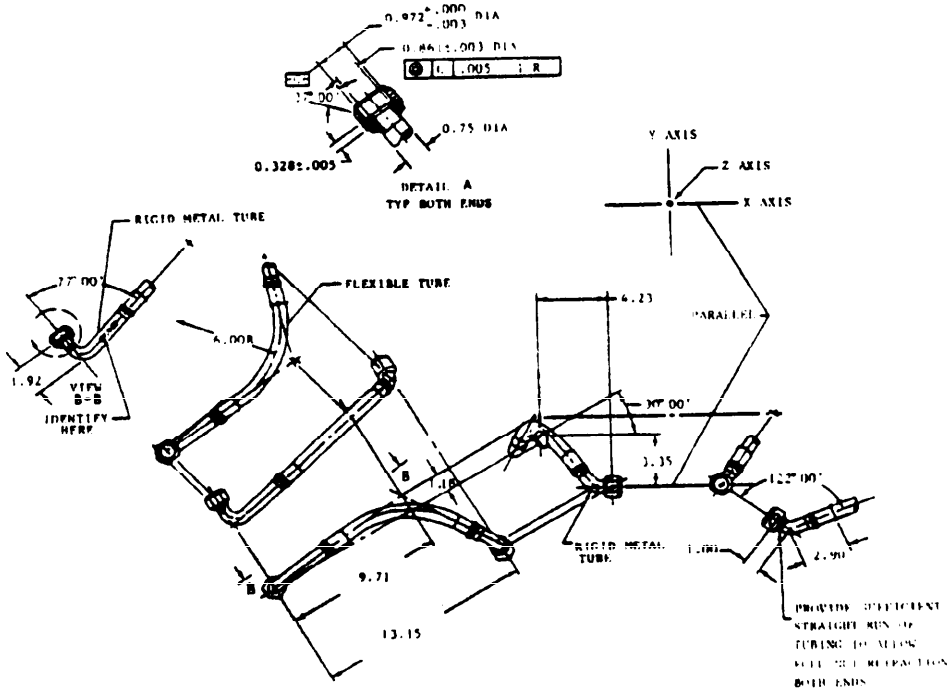


NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 3. Hose assembly type III.

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NOTES:

1. All dimensions are in inches.
2. Metric equivalents shall be acceptable.

FIGURE 4. Hose assembly type IV.

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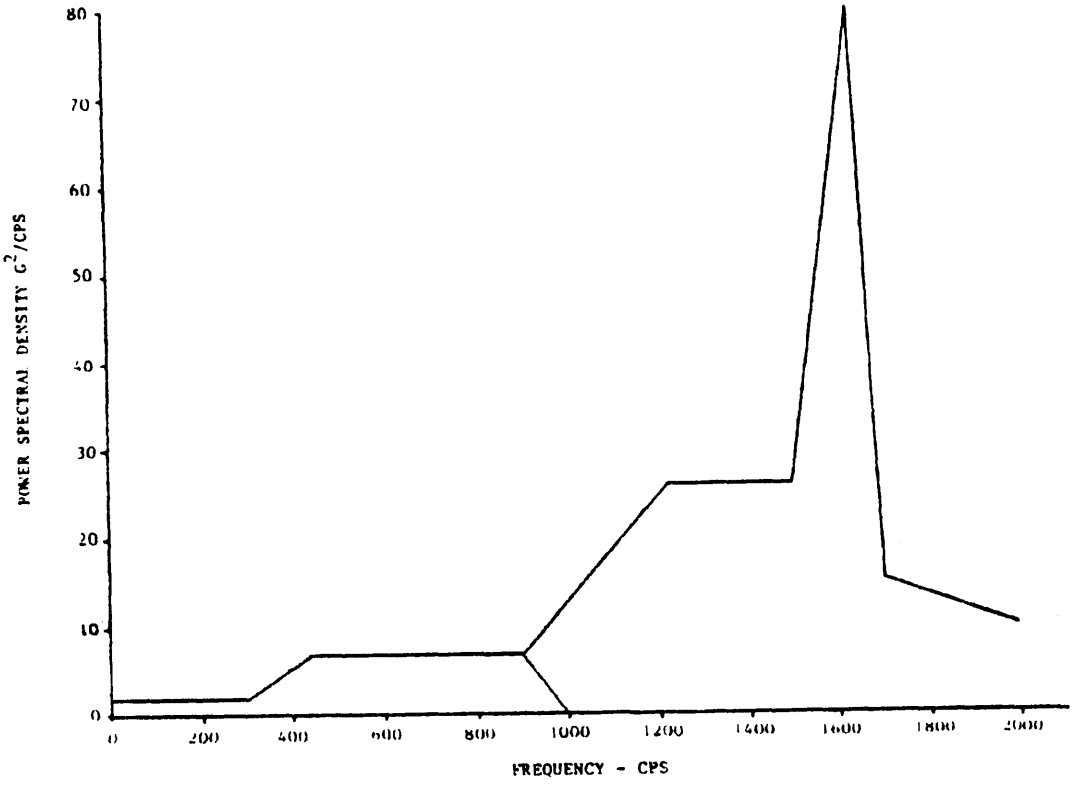


FIGURE 5. Random vibration spectrum.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-47305(MI)	HOSE ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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